

SECTION C

10 (**short**)

(a) Derive the expression $C = \epsilon A/d$ for a parallel plate capacitor, defining each of the symbols in the expression. [4]

(b) As shown in Fig. 6, a capacitor is made with a solid-foam dielectric of thickness d and relative permittivity $\epsilon_r = 1$. The fixed top electrode is a sheet of metal glued to the dielectric. The bottom electrode is a metal sheet which is attracted to the bottom face of the dielectric by the electric field, applied by thin, light and flexible wires. Find an expression for the minimum voltage V that must be applied if the metal sheet, of density ρ per unit area, is not to fall from the dielectric. [6]

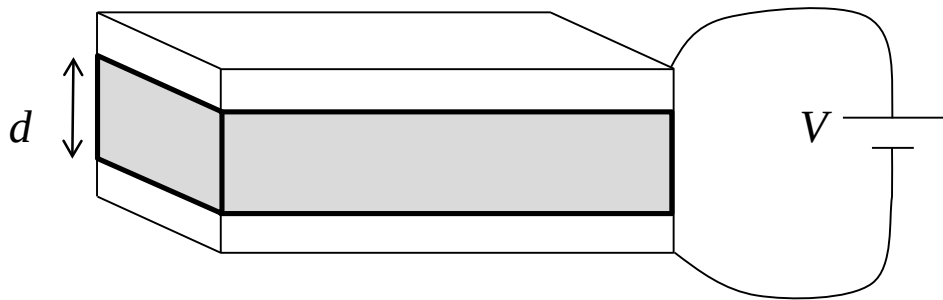


Fig. 6